Experiment 10: Structures

Objective: To learn how to define, declare, initialize, and manipulate structures in C, and to understand how structures can be used to store and process related data.

Questions:

1. **Write a program to define a `Student` structure with name, roll number, and marks. Input data for 3 students and print the student with the highest marks.**
   Sample Input:
   ```
   Student 1: John, 101, 85
   Student 2: Alice, 102, 92
   Student 3: Bob, 103, 78
   ```
   Sample Output:
   ```
   Topper: Alice (Roll No: 102) with 92 marks
   ```

2. **Create a structure `Time` with hour, min, sec. Write a function to add two Time variables and return the result.**
   Sample Input:
   ```
   Time 1: 2 hrs 45 mins 50 secs
   Time 2: 1 hrs 20 mins 30 secs
   ```
   Sample Output:
   ```
   Sum: 4 hrs 6 mins 20 secs
   ```

3. **Define a structure `Complex` with real and imaginary parts. Write functions to:**
   a. **Add**
   b. **Subtract**
   c. **Multiply two complex numbers**

   Return the result using structure return.
   Sample Input:
   ```
   Complex 1: 3 + 4i
   Complex 2: 1 + 2i
   ```
   Sample Output:
   ```
   Addition: 4 + 6i
   Subtraction: 2 + 2i
   Multiplication: -5 + 10i
   ```

4. **Define a structure `BankAccount` with account number, name, and balance. Write a function that takes a structure pointer and:**
   a. **Adds 5% interest if balance > 10000**
   b. **Deducts ₹100 penalty if balance < 1000**

   Sample Input:
   ```
   Account 1: 1001, Ravi, ₹12000
   Account 2: 1002, Meena, ₹950
   ```
   Sample Output:
   ```
   Ravi (Account: 1001) Updated Balance: ₹12600.00
   Meena (Account: 1002) Updated Balance: ₹850.00
   ```

5. **Write a program that dynamically allocates memory for an array of structures `Book` using `malloc()`. Each book has a title, author, and price. Input and display the details of `N` books.**
   Sample Input:

```
Book 1: "C Programming", Dennis Ritchie, ₹450
Book 2: "Let Us C", Yashwant Kanetkar, ₹380
```

Sample Output:

```
Book Details:
1. C Programming by Dennis Ritchie – ₹450
2. Let Us C by Yashwant Kanetkar – ₹380
```